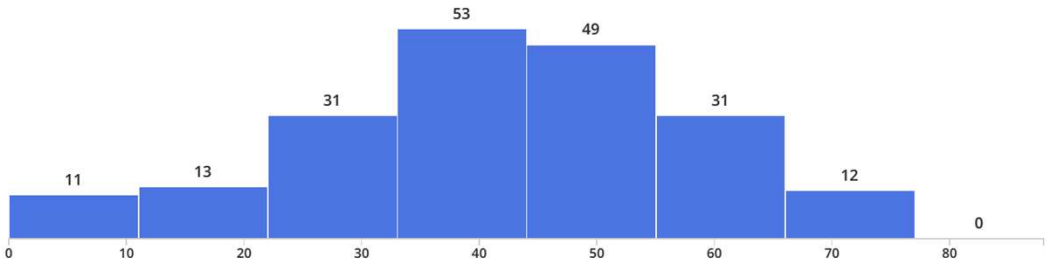
Part B



Minimum	Median	Maximum	Mean	Std Dev
0.0	42.0	73.0	40.95	16.36

3

Part B Instructions

- You may assume any result in the Piazza-provided lecture handouts as “known”, i.e., not to be proved.
- Conversely, you cannot assume results
  - that have not been presented in class or posted in the handouts.
  - that are implicit or explicit in homework solutions
  - Such results are to be derived if needed during the exam.
- All questions require precise and complete mathematical arguments to earn full credit

4

Agenda

- Midsemester Exam
  - Self evaluation
  - Menti
  - Question
  - Answers
    - Piazza: common mistakes, imprecise answers, incorrect answer, partial marking
  - Results
  - Display of answer papers
    - Regrade request process
    - Friday, Students with Low Marks,

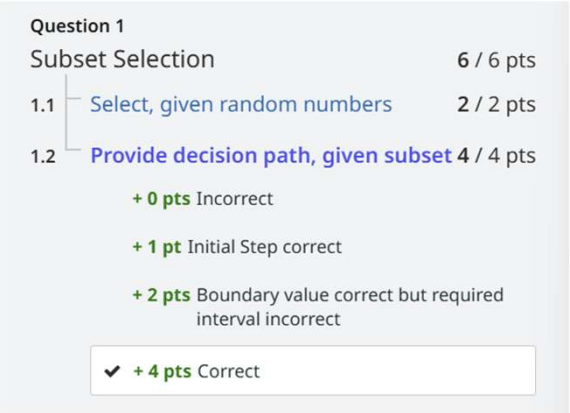
5

Subset Selection

- {B,D,A,E,C,F,G,H}
  - 0.52, 0.57, 0.66, 0.30, 0.80, 0.90, 0.25, 0.70
- Correct subset {D,E,G,H}
  - 4/8, 4/7 ~= 0.571, 3/6, 3/5, 2/4, 2/3
- Decision path for subset {D, E, F, H}
  - [0.50, 1): B Not selected, (0, 4/7): D: Selected, [0.5, 1) A not selected
  - (0, 3/5): E: Selected, [0.5, 1): C: Not Selected, (0, 0.66) F: Selected
  - [0.50, 1): G Not selected, H: Selected (Forced)

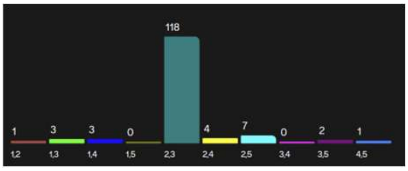
6

Grading Interface



7

Results



1: Subset Selection 6.0 points

1.1: Select, given random numbers 2.0 points 66%

Click to Add Tags

1.2: Provide decision path, given subset **4.0 points** 63%

Click to Add Tags

8

Attendance requirement

- 44 scheduled classes, 80% threshold met implies DX grade cannot be given for lack of attendance
- Identify the binomial parameters
  - Binomial(21, 0.75)

Student	Classes attended
A	16
B	18
C	20

9

Attendance requirement

Student	Classes attended
A	16
B	18
C	20

- 44 scheduled classes, 80% threshold met implies DX grade cannot be given for lack of attendance
- Identify the binomial parameters
  - Binomial(21, 0.75)
- Student A will not meet threshold if he attends 0 classes, or 1 classes, or 2 classes, and so on up to 17 classes since 36-16-2=18
- P(X <= 17) = 0.808 which is greater than the 4% threshold specified

10

Results



11

Red and Blue Balls in Urns

- One of two urns, A or B, is chosen with equal probability. Urn A contains 3 red balls and 1 blue ball, while urn B contains 1 red ball and 3 blue balls. Balls are drawn independently **with replacement** from the chosen urn.
- If the first two balls drawn are red, find the probability that urn A was chosen.
- Suppose that among the first n draws, exactly r are red. Find the condition on r and n under which urn A is more likely than urn B.

12

Results



13

Jenny and IPL franchise

- X and Y are independent normal with the same mean (70) and variance (64)
  - Want to understand S = X+Y and D=X-Y
- Find $\mathbb{E}[e^{uS+vD}]$ and simplify (*state assumptions you make*)
- Find $\mathrm{Cov}(S, D)$
- Find $\Pr(D \geq 8 | S > 150)$. Unjustified assumptions receive no credit

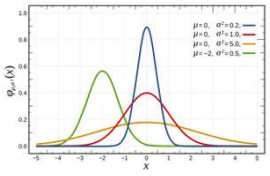
14

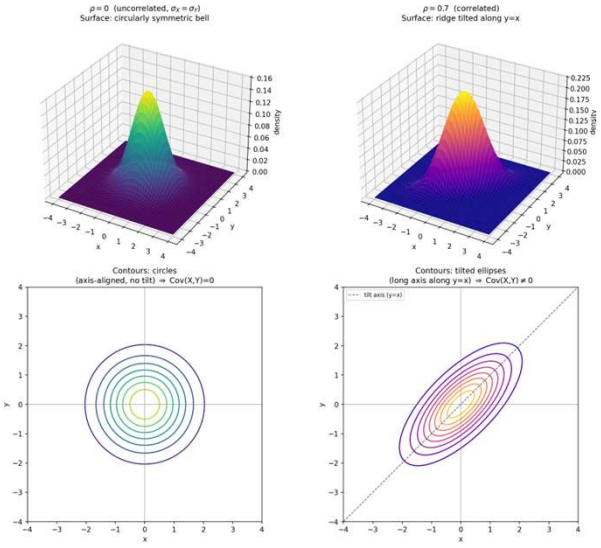
Results

4: IPL	13.0 points		
4.1: Expected Value of e^(uS + vD)	**6.0 points**		65%
Click to Add Tags			
4.2: Cov(S,D)	3.0 points		79%
Click to Add Tags			
4.3: Probability of difference > 8 given sum > 150	4.0 points		0%
Click to Add Tags			

15

Joint distributions of S and D





In both cases, any slice of the "Mexican" hat will give us a nice Gaussian

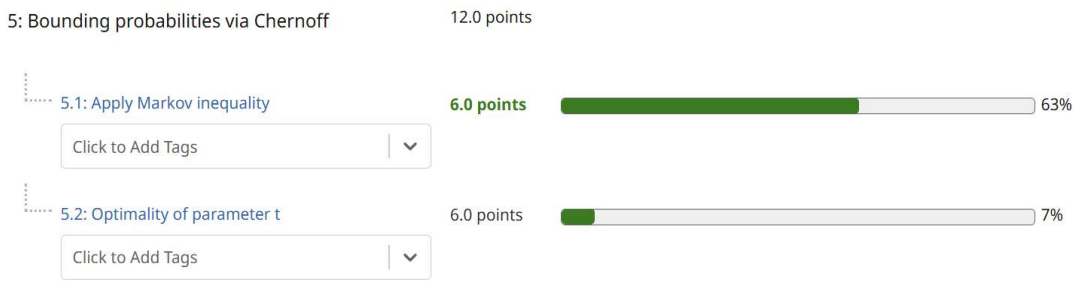
16

Bounding probability of the binomial

- We know the pmf of the binomial, but what about the cdf?
- Simpler question: What is the probability that in *n* trials we get more than *np* successes (e.g. *nq*, with $0 < p < q < 1$)
  - Standard Markov $\Pr(S_n \geq nq) \leq \frac{\mathbb{E}[S_n]}{nq}$
  - With p = 0.2, q=0.5, direct Markov gives 0.4 (no dependance on n)
  - With $np < a < n$ and $q = \frac{a}{n}, p < q < 1$. Show
    - $P(S_n \geq nq) \leq \left[\left(\frac{p}{q}\right)^q \left(\frac{1-p}{1-q}\right)^{1-q}\right]^n$
- This is the tightest Chernoff bound (in the example 0.8^n)

17

Results



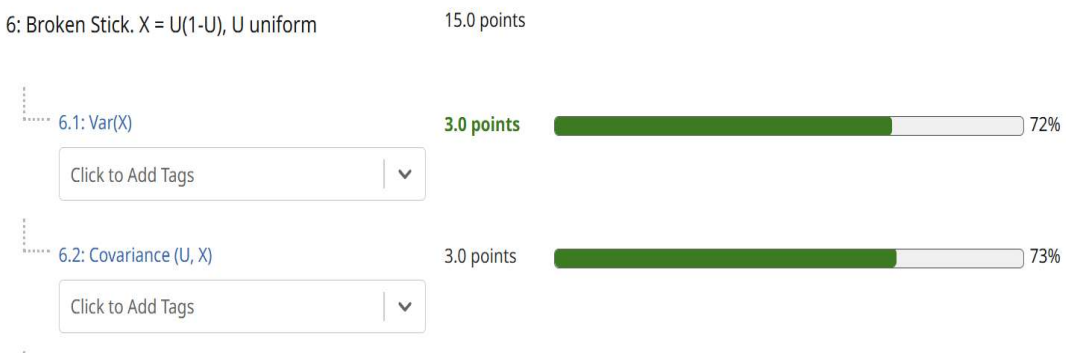
18

Stick

- Stick broken into two: X = U(1-U)
  - Expectation, Variance, Covariance (X, U)

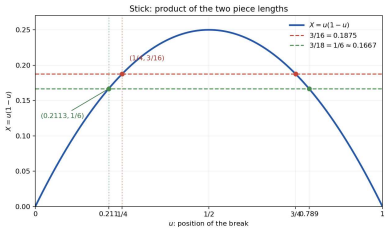
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Results



20

Stick



- Stick broken into two: X = U(1-U)
  - Earlier, covariance = 0 gave us suspicion of independence, now covariance = 0 asks us to check independent or not?
  - Umm X = f(U) that sounds suspiciously like not independent
    - Want $\Pr(X = a|U = u) \stackrel{?}{=} \Pr(X = a)$?
  - We want to look for events that knowing something about X makes U to be constrained, it can't be whatever it wants
    - Argue about the event $\Pr(X \geq 3/16|U < 1/4)$

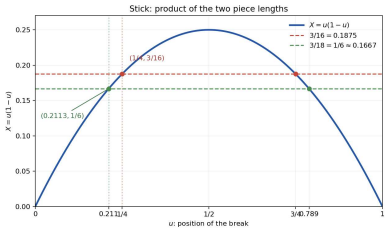
21

Results



22

Stick



- Stick broken into two: X = U(1-U)
  - Earlier, covariance = 0 gave us suspicion of independence, now covariance = 0 asks us to check independent or not?
  - Wait, there is more! show that X and Y are independent
    - Y=1 if U < 0.5
    - This appears to be unexpected!

23

Results

6.4: Independence of X and Y **(only 99% graded)** 6.0 points 10%

Click to Add Tags

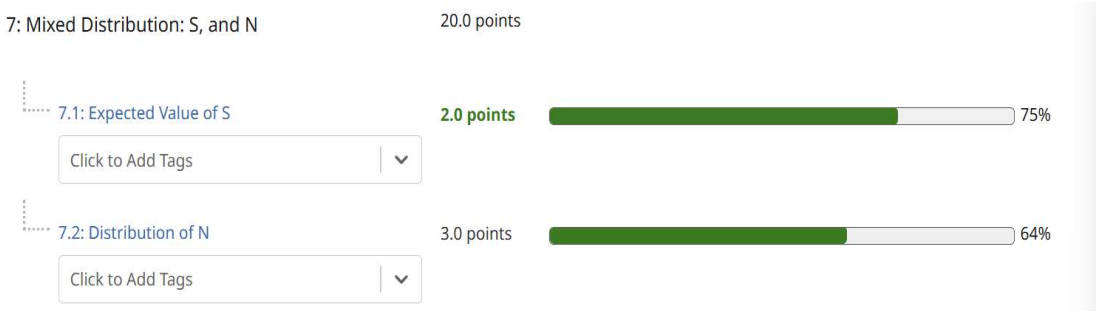
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Mixed Distribution

- S = sum of i.i.d. X_i(continuous, symmetric)
- N counts the number of such $X_i$which is positive
- Find
  - $\mathbb{E}[S]$
  - Distribution of N

25

Results



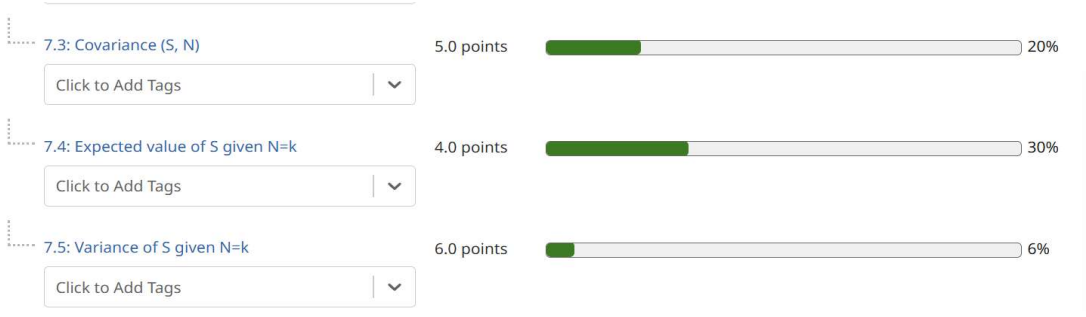
26

Mixed Distribution

- S = sum of i.i.d. X_i(continuous, symmetric)
- N counts the number of such $X_i$which is positive
- Find
  - $\mathrm{Cov}(S, N)$
  - $\mathbb{E}[S \mid \mathrm{N} = \mathrm{k}]$
  - $\mathrm{Var}(S|N = k)$

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Results



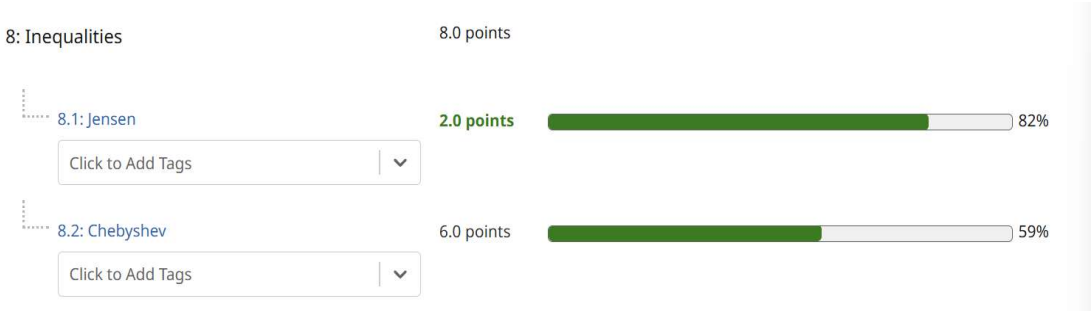
28

Chebychev

- X is random with expectation μ and finite variance
- Jensen's inequality
  - For a convex function, $f\big(\mathbb{E}(x)\big) \leq \mathbb{E}[f(x)]$
- For $0 < \theta < 1$, $\Pr(X \leq \theta\mu) \leq$?

29

Results



30

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